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NLP Lab 17.2 - Sequence-to-Sequence Translation using Hugging Face

Objective

Understand and implement machine translation using pretrained Transformer models.

```
!pip install transformers torch
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.7.1)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.4.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.24.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.12.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (2026.2.25)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (3.11)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (8.3.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (4
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->tran
```

```
from transformers import pipeline
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# French Translator
```

```
model_name_fr = "Helsinki-NLP/opus-mt-en-fr"
tokenizer_fr = AutoTokenizer.from_pretrained(model_name_fr)
model_fr = AutoModelForSeq2SeqLM.from_pretrained(model_name_fr)

def translate_en_to_fr(text):
    inputs = tokenizer_fr(text, return_tensors="pt")
    outputs = model_fr.generate(**inputs)
    return tokenizer_fr.decode(outputs[0], skip_special_tokens=True)

# German Translator
model_name_de = "Helsinki-NLP/opus-mt-en-de"
tokenizer_de = AutoTokenizer.from_pretrained(model_name_de)
model_de = AutoModelForSeq2SeqLM.from_pretrained(model_name_de)

def translate_en_to_de(text):
    inputs = tokenizer_de(text, return_tensors="pt")
    outputs = model_de.generate(**inputs)
    return tokenizer_de.decode(outputs[0], skip_special_tokens=True)
```

```
tokenizer_config.json: 100%  42.0/42.0 [00:00<00:00, 764B/s]

source.spm: 100%  778k/778k [00:00<00:00, 7.30MB/s]

target.spm: 100%  802k/802k [00:00<00:00, 40.9MB/s]

vocab.json:  1.34M/? [00:00<00:00, 35.3MB/s]

/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip install sacremoses.
  warnings.warn("Recommended: pip install sacremoses.")

pytorch_model.bin: 100%  301M/301M [00:03<00:00, 189MB/s]

Loading weights: 100%  258/258 [00:00<00:00, 731.96it/s, Materializing param=model.shared.weight]

model.safetensors: 100%  301M/301M [00:03<00:00, 169MB/s]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but both are present in the checkpoints, s
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but both are present in the checkpoints, s

generation_config.json: 100%  293/293 [00:00<00:00, 17.0kB/s]

config.json:  1.33k/? [00:00<00:00, 93.6kB/s]

tokenizer_config.json: 100%  42.0/42.0 [00:00<00:00, 4.10kB/s]

source.spm: 100%  768k/768k [00:00<00:00, 37.9MB/s]

target.spm: 100%  797k/797k [00:00<00:00, 36.4MB/s]

vocab.json:  1.27M/? [00:00<00:00, 32.1MB/s]

pytorch_model.bin: 100%  298M/298M [00:02<00:00, 177MB/s]

Loading weights: 100%  258/258 [00:00<00:00, 579.00it/s, Materializing param=model.shared.weight]

model.safetensors: 100%  298M/298M [00:05<00:00, 93.0MB/s]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but both are present in the checkpoints, s
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but both are present in the checkpoints, s

generation_config.json: 100%  293/293 [00:00<00:00, 8.53kB/s]
```

```
sentences = [
    "Hello, how are you?",
    "Artificial Intelligence is transforming the world.",
    "I love learning new technologies.",
    "What is your name?",
    "The weather is very pleasant today.",
    "Machine learning models require data.",
    "This is a complex sentence with multiple clauses.",
    "Can you help me with this problem?",
    "Natural language processing is fascinating.",
    "We are preparing for the exam.",
    "He went to the market and bought fruits.",
    "Deep learning uses neural networks.",
    "The system failed due to an unknown error.",
    "Please submit the assignment on time.",
    "Technology is evolving rapidly."
]

sentences
```

```
['Hello, how are you?',
 'Artificial Intelligence is transforming the world.',
 'I love learning new technologies.',
 'What is your name?',
 'The weather is very pleasant today.',
 'Machine learning models require data.',
 'This is a complex sentence with multiple clauses.',
 'Can you help me with this problem?',
 'Natural language processing is fascinating.',
 'We are preparing for the exam.',
 'He went to the market and bought fruits.',
 'Deep learning uses neural networks.',
 'The system failed due to an unknown error.',
 'Please submit the assignment on time.',
 'Technology is evolving rapidly.']
```

```
for s in sentences:
    fr = translate_en_to_fr(s)
    de = translate_en_to_de(s)
    print(f"Original: {s}")
    print(f"French: {fr}")
    print(f"German: {de}")
    print("-"*50)
```

```
Original: Hello, how are you?
French: Bonjour, comment allez-vous ?
German: Hallo, wie geht's?
--------------------------------------------------
Original: Artificial Intelligence is transforming the world.
French: L'intelligence artificielle transforme le monde.
German: Künstliche Intelligenz verwandelt die Welt.
--------------------------------------------------
Original: I love learning new technologies.
French: J'adore apprendre les nouvelles technologies.
German: Ich liebe es, neue Technologien zu lernen.
--------------------------------------------------
Original: What is your name?
French: Quel est votre nom ?
```

```
German: Wie heißt du?
--------------------------------------------------
Original: The weather is very pleasant today.
French: Le temps est très agréable aujourd'hui.
German: Das Wetter ist heute sehr angenehm.
--------------------------------------------------
Original: Machine learning models require data.
French: Les modèles d'apprentissage automatique nécessitent des données.
German: Maschinelle Lernmodelle erfordern Daten.
--------------------------------------------------
Original: This is a complex sentence with multiple clauses.
French: Il s'agit d'une phrase complexe avec plusieurs clauses.
German: Dies ist ein komplexer Satz mit mehreren Klauseln.
--------------------------------------------------
Original: Can you help me with this problem?
French: Pouvez-vous m'aider avec ce problème ?
German: Können Sie mir bei diesem Problem helfen?
--------------------------------------------------
Original: Natural language processing is fascinating.
French: Le traitement naturel du langage est fascinant.
German: Natürliche Sprachverarbeitung ist faszinierend.
--------------------------------------------------
Original: We are preparing for the exam.
French: Nous nous préparons à l'examen.
German: Wir bereiten uns auf die Prüfung vor.
--------------------------------------------------
Original: He went to the market and bought fruits.
French: Il est allé au marché et a acheté des fruits.
German: Er ging auf den Markt und kaufte Früchte.
--------------------------------------------------
Original: Deep learning uses neural networks.
French: L'apprentissage profond utilise des réseaux neuronaux.
German: Deep Learning nutzt neuronale Netzwerke.
--------------------------------------------------
Original: The system failed due to an unknown error.
French: Le système a échoué en raison d'une erreur inconnue.
German: Das System ist aufgrund eines unbekannten Fehlers fehlgeschlagen.
--------------------------------------------------
Original: Please submit the assignment on time.
French: Veuillez soumettre l'affectation à temps.
German: Bitte reichen Sie den Auftrag rechtzeitig ein.
--------------------------------------------------
Original: Technology is evolving rapidly.
French: La technologie évolue rapidement.
```

Step 5: Analysis

- The translations are generally accurate.
- Grammar is mostly correct.
- Complex sentences may slightly lose meaning.
- Technical terms are sometimes preserved.
- Different languages show structural differences.

Conclusion:

Pretrained Transformer models provide efficient and accurate translation without training.